

EM FORMULAS

①

UNIT-1 Fundamental of Mechanics

1. Parallelogram law

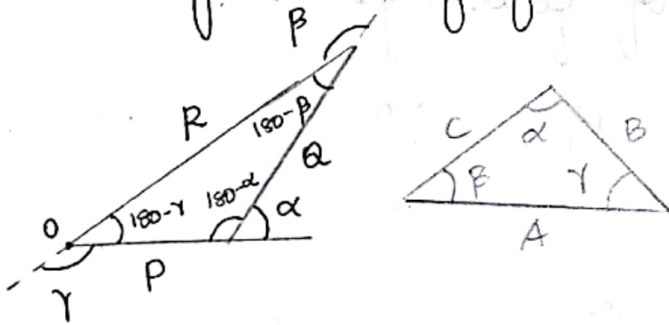
Resultant (R)

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

Direction of resultant (α)

$$\alpha = \tan^{-1} \left(\frac{F_2 \sin \theta}{F_1 + F_2 \cos \theta} \right)$$

2. Triangle law of force.



$$A^2 = B^2 + C^2 - 2BC \cos \alpha$$

$$B^2 = A^2 + C^2 - 2AC \cos \beta$$

$$C^2 = A^2 + B^2 - 2AB \cos \gamma$$

$$\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{C}{\sin \gamma}$$

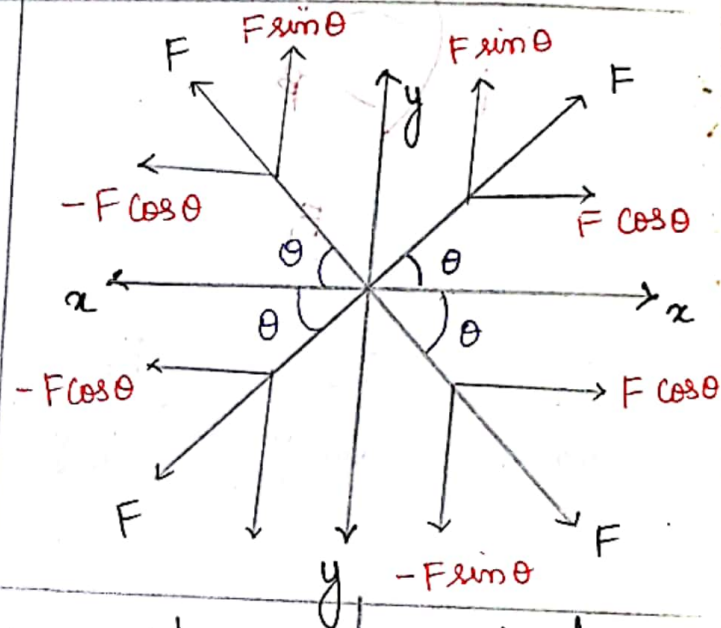
3. Resolution of forces.

Resultant

$$R = \sqrt{\sum F_x^2 + \sum F_y^2}$$

Direction

$$\alpha = \tan^{-1} \left(\frac{\sum F_y}{\sum F_x} \right)$$



Horizontal Component

Force $\times \cos \theta$ $\rightarrow +ve$ $\leftarrow -ve$

Vertical Component

Force $\times \sin \theta$ $\uparrow +ve$ $\downarrow -ve$

4. Equilibrium condition.

$$i) \sum F_x = 0$$

$$ii) \sum F_y = 0$$

$$\sum F_x + \sum F_y = 0$$

Units :

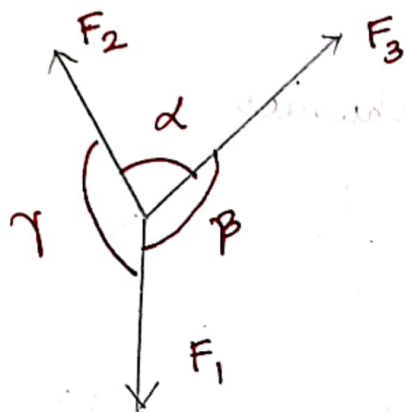
Resultant (R) = N (Newton)

Direction (α) = $^\circ$ (degree)

Force (F) = N (newton)

Moment (M) = N.m (Newton metre)

5. Lami's theorem.



$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

6. Varignon's theorem. (2)

$$\Sigma M = R \times x$$

Moment = Force $\times$ perpendicular distance

R - Resultant (N)

x - location (m)

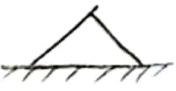
7. Couple.

clockwise
(+ve)

anticlockwise
(-ve)

Unit - 2.

Practical application of force systems.

Type of support	Support reaction	No. of unknowns.
(1) Roller support 	Vertical (V) $\uparrow$	01
(2) Hinged support 	Horizontal $\rightarrow$ Vertical $\uparrow$	02
(3) Fixed support 	Horizontal (H) $\rightarrow$ Vertical (V) $\uparrow$ Moment (M) $\curvearrowright$	03

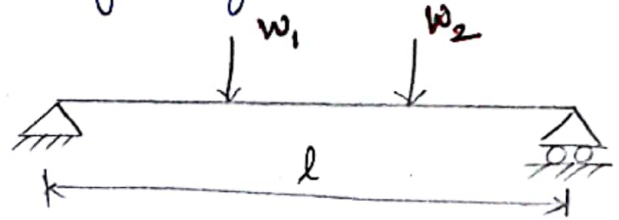
Equilibrium conditions

$$\sum H = 0$$

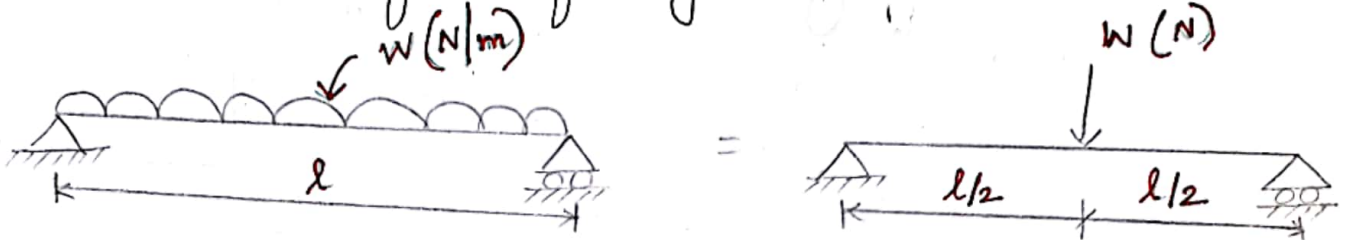
$$\sum V = 0$$

$$\sum M = 0$$

Types of load : (i) point load



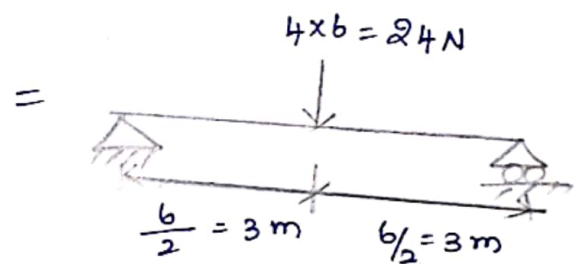
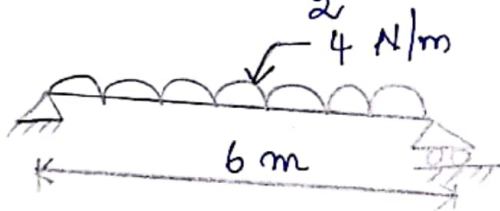
(ii) Conversion of uniformly distributed load (udl) to pt-load



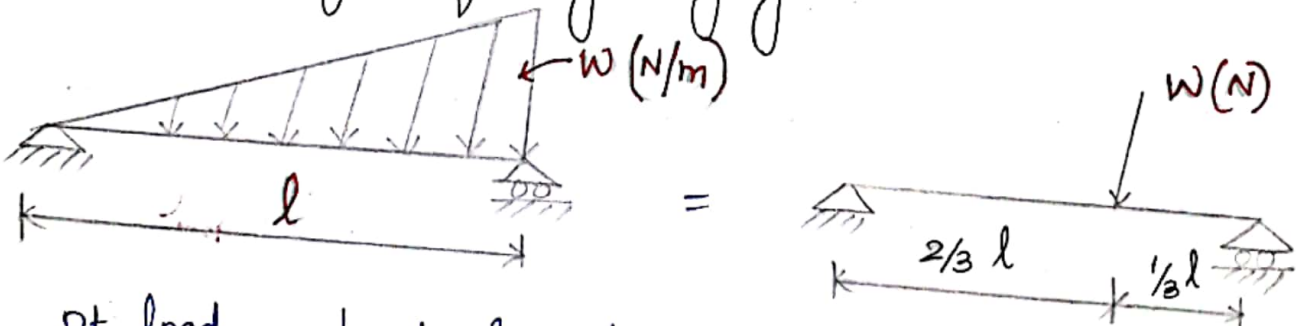
$$\text{pt. load} = \text{udl} \times \text{distance (l)} = w \times l$$

$$\text{distance} = \frac{l}{2}$$

Eg:



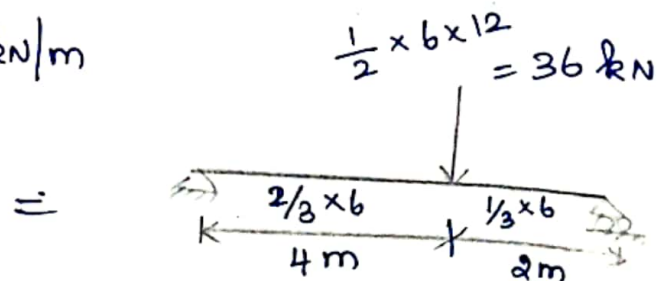
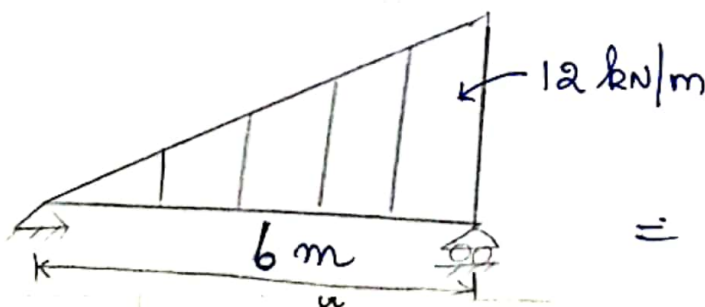
(iii) Conversion of uniformly varying load (uvl) to point load.



$$\text{pt. load} = \frac{1}{2} \times b \times h = \frac{1}{2} \times l \times w$$

$$\text{distance} = \frac{1}{3} l \text{ and } \frac{2}{3} l$$

Eg:



Method of joint:

⇒ Tensile force (+ve) $\longleftrightarrow T$

⇒ Compressive force (-ve) $\rightarrow C \leftarrow$

Calculation :

Step 1: Draw free body diagram.

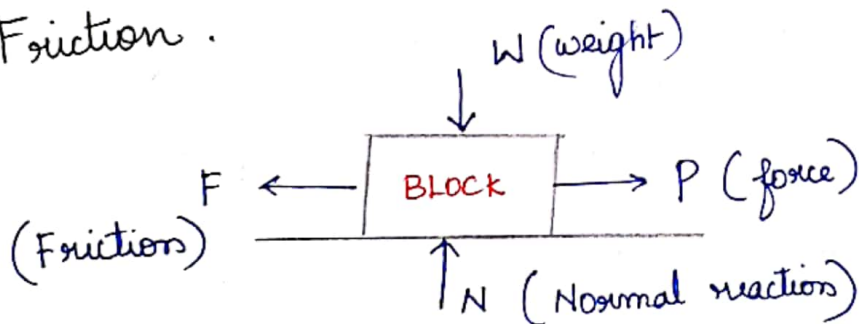
Step 2: Calculate support reaction using Equilibrium Condition (ie $\Sigma H=0, \Sigma V=0, \Sigma M=0$)

Step 3: Analyse using Method of joint

Step 4: Tabulate the results

Step 5: Draw the result design.

Friction.

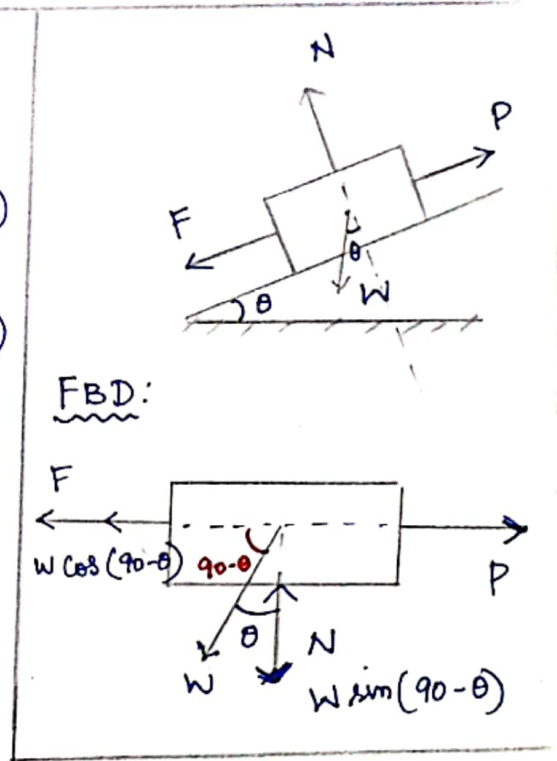


Co-efficient of friction (μ) = $\frac{F}{N}$.

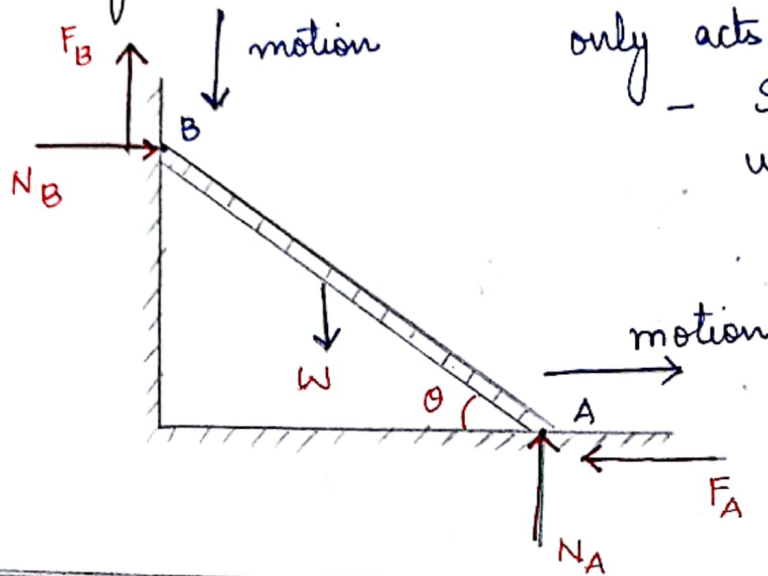
μ - No unit.

- Hint :
- * motion opposite friction
 - * Weight opposite reaction

- * Surface smooth $\Rightarrow$ friction will not be there.
- * Friction occurs only on rough surface.
- * Reaction will always acts perpendicular to the surface



Ladder friction :

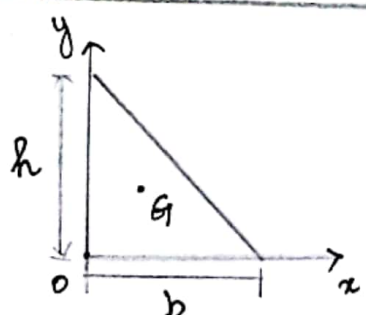
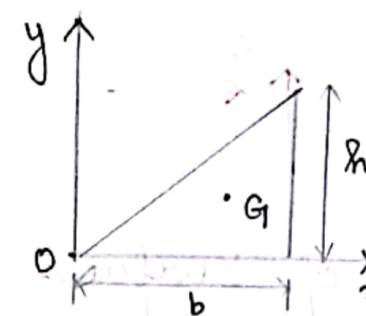
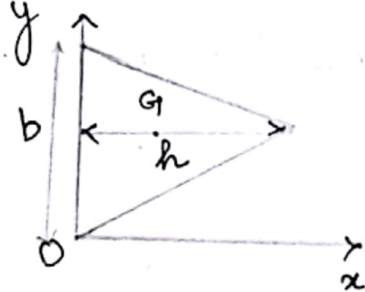
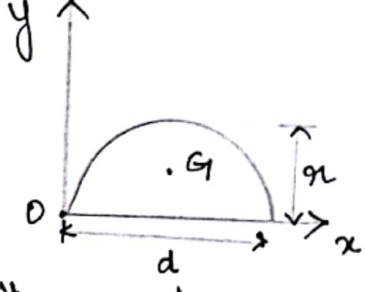
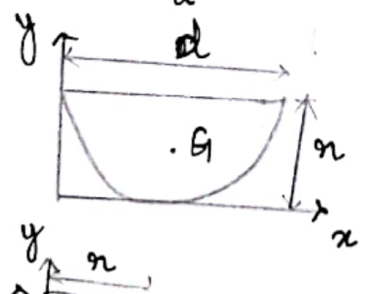
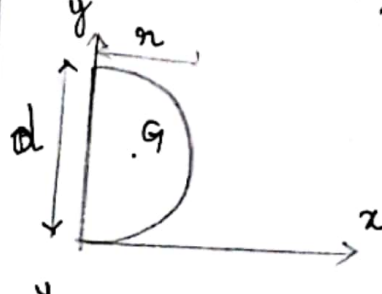
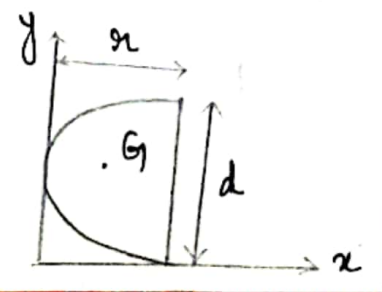


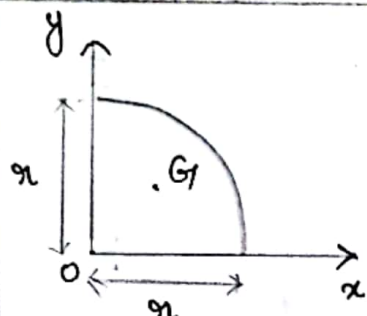
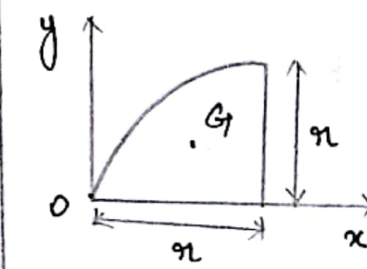
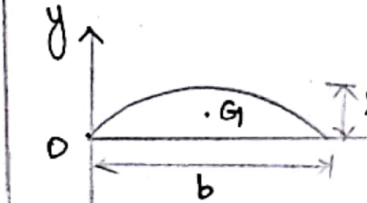
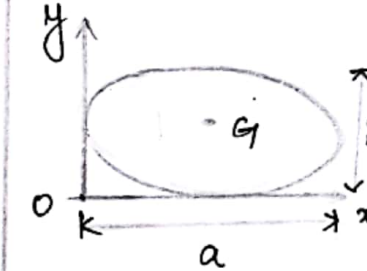
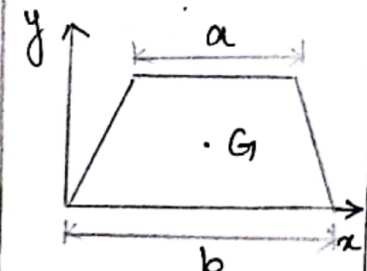
Hint : Weight (w) of the ladder only acts at the centre. ⑤
 - Solve the ladder problems using 4 conditions
 $\Sigma H=0$, $\Sigma V=0$
 $\Sigma M=0$, $\mu = \frac{F}{N}$.

UNIT-3 Properties of Surfaces.

CENTROID (G) (OR) CENTRE OF GRAVITY

Sl. No.	Name	Shape	$\bar{x}$	$\bar{y}$	Area
1.	Square		$\frac{a}{2}$	$\frac{a}{2}$	a^2
2.	Rectangle		$\frac{l}{2}$	$b/2$	lb
3.	Triangle		$\frac{b}{2}$	$\frac{h}{3}$	$\frac{1}{2}bh$
4.	Circle		$\frac{d}{2}$ (or) r	$\frac{d}{2}$ (or) r	πr^2

Sl. NO.	Name	Shape	$\bar{x}$	$\bar{y}$	Area (6)
5.	Right angled triangle		$\frac{b}{3}$	$\frac{h}{3}$	$\frac{1}{2}bh$
6.	Right-angled triangle		$\frac{2b}{3}$	$\frac{h}{3}$	$\frac{1}{2}bh$
7.	Triangle		$\frac{h}{3}$	$\frac{b}{2}$	$\frac{1}{2}bh$
8.	Semi-circle		r	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{2}$
9.	Semi-circle		r	$r - \frac{4r}{3\pi}$	$\frac{\pi r^2}{2}$
10.	Semi-circle		$\frac{4r}{3\pi}$	r	$\frac{\pi r^2}{2}$
11.	Semi-circle		$r - \frac{4r}{3\pi}$	r	$\frac{\pi r^2}{2}$

Sl. No.	Name	Shape	$\bar{x}$	$\bar{y}$	Area (1)
12.	Quadrant		$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{1}{4} \pi r^2$
13.	Quadrant		$r - \frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{1}{4} \pi r^2$
14.	Parabola		$\frac{b}{2}$	$\frac{2}{5} h$	$\frac{2}{3} b h$
15.	Ellipse		$\frac{a}{2}$	$\frac{b}{2}$	$\frac{\pi a b}{4}$
16.	Trapezium		$\frac{b}{2}$	$\left[\frac{b+2a}{b+a} \right] \frac{h}{3}$	$\frac{1}{2} (a+b) h$

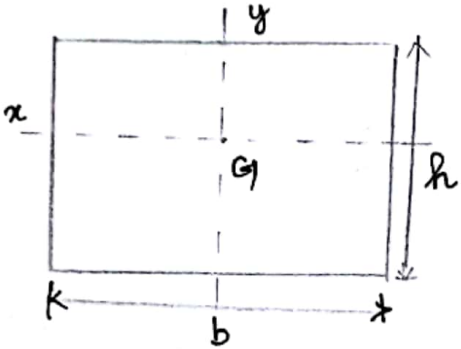
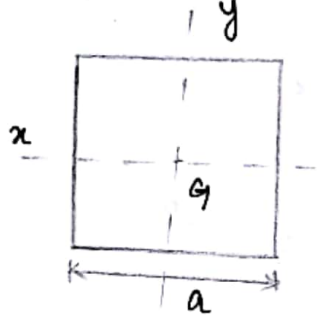
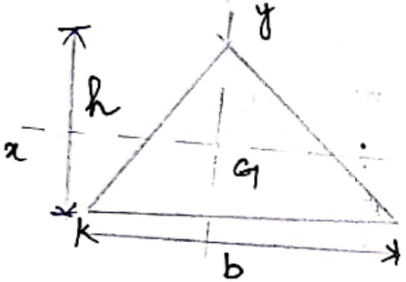
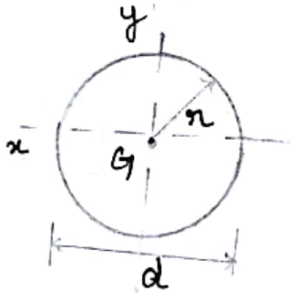
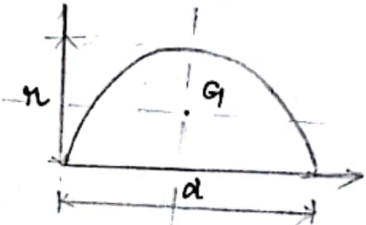
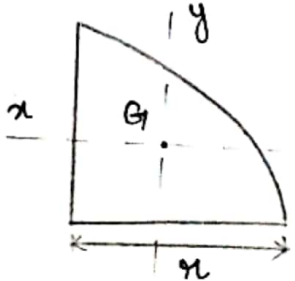
Centroid (or) Centre of gravity (G) :

$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3 + \dots + a_n x_n}{a_1 + a_2 + a_3 + \dots + a_n} = \frac{\sum a x}{\sum a}$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + \dots + a_n y_n}{a_1 + a_2 + \dots + a_n} = \frac{\sum a y}{\sum a}$$

Moment of Inertia (I)

8

Sl. No.	Name	Shape	I_{xx}	I_{yy}
1.	Rectangle		$\frac{bh^3}{12}$	$\frac{hb^3}{12}$
2.	Square		$\frac{a^4}{12}$	$\frac{a^4}{12}$
3.	Triangle		$\frac{bh^3}{36}$	$\frac{hb^3}{48}$
4.	Circle		$\frac{\pi r^4}{4}$	$\frac{\pi r^4}{4}$
5.	Semi-circle		$0.108 r^4$	$0.39 r^4$
6.	Quadrant		$0.055 r^4$	$0.055 r^4$

Moment of inertia about x-axis

$$I_{xx} = (I_{xx})_i + A(\bar{y} - y)^2$$

Moment of inertia about y-axis

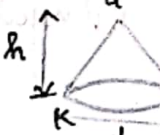
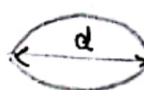
$$I_{yy} = (I_{yy})_i + A(\bar{x} - x)^2$$

Polar moment of inertia (I_p) = $I_{xx} + I_{yy}$

Radius of gyration $r_x = \sqrt{\frac{I_{xx}}{A}}$

$$r_y = \sqrt{\frac{I_{yy}}{A}}$$

Centre of gravity of solid figures

Sl. No.	Name	Shape	$\bar{x}$	$\bar{y}$	Volume
1.	Cylinder		r	$h/2$	$\pi r^2 h$
2.	Cone		r	$h/4$	$\frac{1}{3} \pi r^2 h$
3.	Sphere		r	r	$\frac{4}{3} \pi r^3$
4.	Hemisphere		r	$\frac{3r}{8}$	$\frac{2}{3} \pi r^3$

Centroid (or) centre of gravity of solid figures

$$\bar{x} = \frac{V_1 x_1 + V_2 x_2 + V_3 x_3 + \dots + V_n x_n}{V_1 + V_2 + V_3 + \dots + V_n} = \frac{\sum V x}{\sum V}$$

$$\bar{y} = \frac{V_1 y_1 + V_2 y_2 + \dots + V_n y_n}{V_1 + V_2 + V_3 + \dots + V_n} = \frac{\sum V y}{\sum V}$$

UNIT-4/5 Kinematics and kinetics of particles and Rigid bodies (10)

Sl. No.	POINT OF COMPARISON	LINEAR MOTION	ANGULAR MOTION
1.	Displacement	s (m)	θ (rad)
2.	Initial velocity	u (m/s)	ω_0 (rad/s)
3.	Final velocity	v (m/s)	ω (rad/s)
4.	Acceleration	a (m/s ²)	α (rad/s ²)
5.	Retardation (or) deceleration	$-a$ (m/s ²)	$-\alpha$ (rad/s ²)
6.	Time	t (sec)	t (sec or min)

Equations of Motion.

1. Linear motion	Angular motion.
1. $S = vt$	i) $\theta = \omega t$
2. $v = u + at$	ii) $\omega = \omega_0 + \alpha t$
3. $v^2 = u^2 + 2as$	iii) $\omega^2 = \omega_0^2 + 2\alpha s$
4. $S = ut + \frac{1}{2}at^2$	iv) $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
5. $S_n = u + \frac{a}{2}(2n-1)$	v) $\theta_n = \omega_0 + \frac{\alpha}{2}(2n-1)$

Hint: $\rightarrow$ When the particle starts from rest $u=0$
 $\rightarrow$ When the particle comes to rest $v=0$

$$g = 9.81 \text{ m/s}^2$$

$\rightarrow$ When the object is thrown upward $v=0$

$\rightarrow$ When the object is thrown downward $u=0$

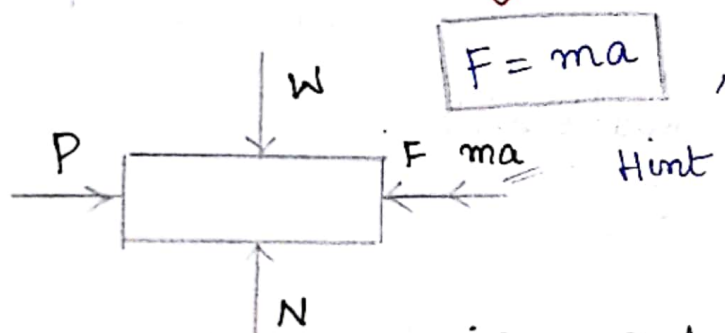
↓ DOWNWARD MOTION (+ve)
⇒ towards the gravity

↑ UPWARD MOTION (-ve)
⇒ against the gravity. (11)

(i) $v = u + gt$
 (ii) $v^2 = u^2 + 2gh$
 (iii) $h = ut + \frac{1}{2}gt^2$
 (iv) $h_n = u + \frac{g}{2}(2n-1)$

i) $v = u - gt$
 ii) $v^2 = u^2 - 2gh$
 iii) $h = ut - \frac{1}{2}gt^2$
 iv) $h_n = u - \frac{g}{2}(2n-1)$

D'Alembert's principle (Newton's 2nd law)



ma - Inertia force.

Hint: Motion opposite Inertia force (ma) will act.

Conversion of weight (w) to mass (m) $\div 9.81$
 " mass (m) to weight (w) $\times 9.81$

Eg: weight = 10 N

mass = $\frac{10}{9.81} = 1.01 \text{ kg}$

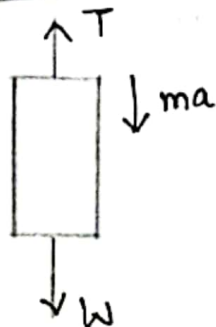
mass = 10 kg

weight = $10 \times 9.81 = 98.1 \text{ N}$

Lift motion

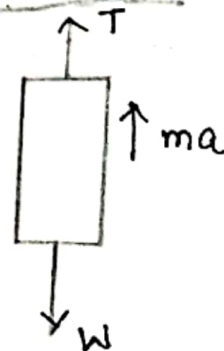
Lift moving upward:

$T - W = ma$



Lift moving downward:

$W - T = ma$



Work Energy principle

(12)

$$\text{Linear motion: } \Sigma F \times s = \frac{m}{2} (v^2 - u^2)$$

$$\text{Angular motion: } \Sigma M_G \times \theta = \frac{I_G}{2} (\omega^2 - \omega_0^2)$$

Impulse momentum Equation:

$$\text{Linear Motion: } \Sigma F \times t = m(v - u)$$

$$\text{Angular motion: } \Sigma M_G \times t = I_G (\omega - \omega_0)$$

$$I_G = \frac{m r^2}{2} \text{ (or) } m k^2$$

r - radius of pulley
 k - radius of gyration

Linear and angular relationship.

$$\omega_0 = \frac{u}{r}, \quad \omega = \frac{v}{r}, \quad \alpha = \frac{a}{r}, \quad \theta = \frac{s}{r}$$

$$\omega = \frac{2\pi N}{60}, \quad N - \text{No. of revolutions (rpm)}$$

$$\text{No. of revolution (N)} = \frac{\theta}{2\pi} \quad (\text{i.e. } \theta = 2N\pi)$$

Chasle's theorem (General plane motion)

= Translation + Rotation

$$\Sigma F \times s = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2$$

Differentiation & Integration (Mathematical Expressions)

$$v = \frac{ds}{dt}, \quad a = \frac{dv}{dt} \text{ (or) } \frac{d^2s}{dt^2}$$

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt} \text{ (or) } \frac{d^2\theta}{dt^2}$$

$$v = \int a \cdot dt; \quad s = \int v \cdot dt$$

$$\omega = \int \alpha \cdot dt, \quad \theta = \int \omega \cdot dt$$

$$a = v \cdot \frac{dv}{ds}$$

Method of Collision / Impact of elastic bodies ^{or elasticity}

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

e - Co-efficient of restitution.
(No. unit)
 $e = 1 \rightarrow$ Impact elastic
 $e = 0 \rightarrow$ " plastic

Law of Conservation of momentum

(13)

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

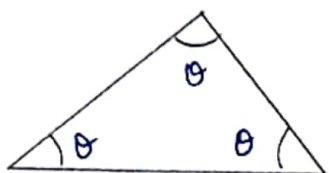
$$\text{Potential Energy} = mgh$$

Loss in Kinetic Energy (K.E) = Total K.E before impact - Total K.E after impact.

$$\text{Loss in K.E} = \left[\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \right] + \left[\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \right]$$

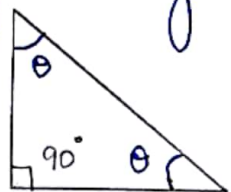
Hint:

(i)



Sum of triangle = 180° .

(ii) Right-angle triangle



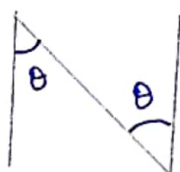
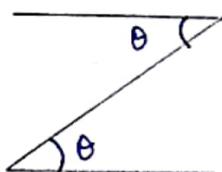
$\Rightarrow$ To find θ , use the below relations:

$$\sin \theta = \frac{\text{opp.}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

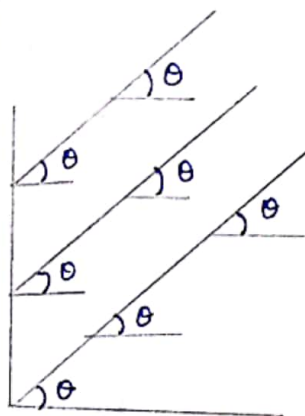
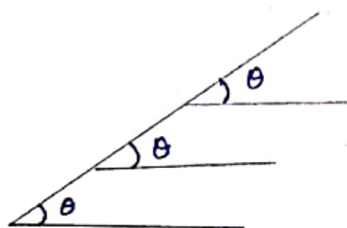
$$\tan \theta = \frac{\text{opp.}}{\text{adj}}$$

(iii)



$\Rightarrow$ Alternate angle are Equal

(iv)



Angle lying on the same slopes are Equal.